Hybrid 8x8 ultrasonic AESA: 64 transceiver channels (CH01..CH64, one sheet reused 64 times), 8 MEMS mics, 8 ADCs. Source of truth: this KiCad hierarchy; skidl/channel.py is the reference spec.

CH01
DRV1 DRV_OUT RX_OUT RX1

CH02
DRV2 DRV_OUT RX_OUT RX2

CH03
DRV3 DRV_OUT RX_OUT RX3

CH04
DRV4 DRV_OUT RX_OUT RX4

CH05
DRV5 DRV_OUT RX_OUT RX5

CH06
DRV6 DRV_OUT RX_OUT RX6

CH07
DRV7 DRV_OUT RX_OUT RX7

CH08
DRV8 DRV_OUT RX_OUT RX8

CH09
DRV9 DRV_OUT RX_OUT RX9

CH10
DRV10 DRV_OUT RX_OUT RX10

CH11
DRV11 DRV_OUT RX_OUT RX11

CH12
DRV12 DRV_OUT RX_OUT RX12

CH13
DRV13 DRV_OUT RX_OUT RX13

CH14
DRV14 DRV_OUT RX_OUT RX14

CH15
DRV15 DRV_OUT RX_OUT RX15

CH16
DRV16 DRV_OUT RX_OUT RX16

CH17
DRV17 DRV_OUT RX_OUT RX17

CH18
DRV18 DRV_OUT RX_OUT RX18

CH19
DRV19 DRV_OUT RX_OUT RX19

CH20
DRV20 DRV_OUT RX_OUT RX20

CH21
DRV21 DRV_OUT RX_OUT RX21

CH22
DRV22 DRV_OUT RX_OUT RX22

CH23
DRV23 DRV_OUT RX_OUT RX23

CH24
DRV24 DRV_OUT RX_OUT RX24

CH25
DRV25 DRV_OUT RX_OUT RX25

CH26
DRV26 DRV_OUT RX_OUT RX26

CH27
DRV27 DRV_OUT RX_OUT RX27

CH28
DRV28 DRV_OUT RX_OUT RX28

CH29
DRV29 DRV_OUT RX_OUT RX29

CH30
DRV30 DRV_OUT RX_OUT RX30

CH31
DRV31 DRV_OUT RX_OUT RX31

CH32
DRV32 DRV_OUT RX_OUT RX32

CH33
DRV33 DRV_OUT RX_OUT RX33

CH34
DRV34 DRV_OUT RX_OUT RX34

CH35
DRV35 DRV_OUT RX_OUT RX35

CH36
DRV36 DRV_OUT RX_OUT RX36

CH37
DRV37 DRV_OUT RX_OUT RX37

CH38
DRV38 DRV_OUT RX_OUT RX38

CH39
DRV39 DRV_OUT RX_OUT RX39

CH40
DRV40 DRV_OUT RX_OUT RX40

CH41
DRV41 DRV_OUT RX_OUT RX41

CH42
DRV42 DRV_OUT RX_OUT RX42

CH43
DRV43 DRV_OUT RX_OUT RX43

CH44
DRV44 DRV_OUT RX_OUT RX44

CH45
DRV45 DRV_OUT RX_OUT RX45

CH46
DRV46 DRV_OUT RX_OUT RX46

CH47
DRV47 DRV_OUT RX_OUT RX47

CH48
DRV48 DRV_OUT RX_OUT RX48

CH49
DRV49 DRV_OUT RX_OUT RX49

CH50
DRV50 DRV_OUT RX_OUT RX50

CH51
DRV51 DRV_OUT RX_OUT RX51

CH52
DRV52 DRV_OUT RX_OUT RX52

CH53
DRV53 DRV_OUT RX_OUT RX53

CH54
DRV54 DRV_OUT RX_OUT RX54

CH55
DRV55 DRV_OUT RX_OUT RX55

CH56
DRV56 DRV_OUT RX_OUT RX56

CH57
DRV57 DRV_OUT RX_OUT RX57

CH58
DRV58 DRV_OUT RX_OUT RX58

CH59
DRV59 DRV_OUT RX_OUT RX59

CH60
DRV60 DRV_OUT RX_OUT RX60

CH61
DRV61 DRV_OUT RX_OUT RX61

CH62
DRV62 DRV_OUT RX_OUT RX62

CH63
DRV63 DRV_OUT RX_OUT RX63

CH64
DRV64 DRV_OUT RX_OUT RX64

ADC1
RX1 IN1 SCLK ADC_SCLK_A
RX2 IN2 CONVST ADC_CONVST_A
RX3 IN3 DOUTA ADC_DOUTA1
RX4 IN4 DOUTB ADC_DOUTB1
RX5 IN5 BUSY ADC_BUSY
RX6 IN6
RX7 IN7
RX8 IN8

ADC2
RX9 IN1 SCLK ADC_SCLK_A
RX10 IN2 CONVST ADC_CONVST_A
RX11 IN3 DOUTA ADC_DOUTA2
RX12 IN4 DOUTB ADC_DOUTB2
RX13 IN5 BUSY
RX14 IN6
RX15 IN7
RX16 IN8

ADC3
RX17 IN1 SCLK ADC_SCLK_A
RX18 IN2 CONVST ADC_CONVST_A
RX19 IN3 DOUTA ADC_DOUTA3
RX20 IN4 DOUTB ADC_DOUTB3
RX21 IN5 BUSY
RX22 IN6
RX23 IN7
RX24 IN8

ADC4
RX25 IN1 SCLK ADC_SCLK_A
RX26 IN2 CONVST ADC_CONVST_A
RX27 IN3 DOUTA ADC_DOUTA4
RX28 IN4 DOUTB ADC_DOUTB4
RX29 IN5 BUSY
RX30 IN6
RX31 IN7
RX32 IN8

ADC5
RX33 IN1 SCLK ADC_SCLK_B
RX34 IN2 CONVST ADC_CONVST_B
RX35 IN3 DOUTA ADC_DOUTA5
RX36 IN4 DOUTB ADC_DOUTB5
RX37 IN5 BUSY
RX38 IN6
RX39 IN7
RX40 IN8

ADC6
RX41 IN1 SCLK ADC_SCLK_B
RX42 IN2 CONVST ADC_CONVST_B
RX43 IN3 DOUTA ADC_DOUTA6
RX44 IN4 DOUTB ADC_DOUTB6
RX45 IN5 BUSY
RX46 IN6
RX47 IN7
RX48 IN8

ADC7
RX49 IN1 SCLK ADC_SCLK_B
RX50 IN2 CONVST ADC_CONVST_B
RX51 IN3 DOUTA ADC_DOUTA7
RX52 IN4 DOUTB ADC_DOUTB7
RX53 IN5 BUSY
RX54 IN6
RX55 IN7
RX56 IN8

ADC8
RX57 IN1 SCLK ADC_SCLK_B
RX58 IN2 CONVST ADC_CONVST_B
RX59 IN3 DOUTA ADC_DOUTA8
RX60 IN4 DOUTB ADC_DOUTB8
RX61 IN5 BUSY
RX62 IN6
RX63 IN7
RX64 IN8

TXG01
TX_SDI0 SDI DRV1 DRV1
TX_SCK_A SCK DRV2 DRV2
TX_LATCH_A LATCH DRV3 DRV3
TX_EN1 EN DRV4 DRV4

TXG02
TX_SDI1 SDI DRV1 DRV5
TX_SCK_A SCK DRV2 DRV6
TX_LATCH_A LATCH DRV3 DRV7
TX_EN1 EN DRV4 DRV8

TXG03
TX_SDI2 SDI DRV1 DRV9
TX_SCK_A SCK DRV2 DRV10
TX_LATCH_A LATCH DRV3 DRV11
TX_EN1 EN DRV4 DRV12

TXG04
TX_SDI3 SDI DRV1 DRV13
TX_SCK_A SCK DRV2 DRV14
TX_LATCH_A LATCH DRV3 DRV15
TX_EN1 EN DRV4 DRV16

TXG05
TX_SDI4 SDI DRV1 DRV17
TX_SCK_B SCK DRV2 DRV18
TX_LATCH_B LATCH DRV3 DRV19
TX_EN2 EN DRV4 DRV20

TXG06
TX_SDI5 SDI DRV1 DRV21
TX_SCK_B SCK DRV2 DRV22
TX_LATCH_B LATCH DRV3 DRV23
TX_EN2 EN DRV4 DRV24

TXG07
TX_SDI6 SDI DRV1 DRV25
TX_SCK_B SCK DRV2 DRV26
TX_LATCH_B LATCH DRV3 DRV27
TX_EN2 EN DRV4 DRV28

TXG08
TX_SDI7 SDI DRV1 DRV29
TX_SCK_B SCK DRV2 DRV30
TX_LATCH_B LATCH DRV3 DRV31
TX_EN2 EN DRV4 DRV32

TXG09
TX_SDI8 SDI DRV1 DRV33
TX_SCK_C SCK DRV2 DRV34
TX_LATCH_C LATCH DRV3 DRV35
TX_EN3 EN DRV4 DRV36

TXG10
TX_SDI9 SDI DRV1 DRV37
TX_SCK_C SCK DRV2 DRV38
TX_LATCH_C LATCH DRV3 DRV39
TX_EN3 EN DRV4 DRV40

TXG11
TX_SDI10 SDI DRV1 DRV41
TX_SCK_C SCK DRV2 DRV42
TX_LATCH_C LATCH DRV3 DRV43
TX_EN3 EN DRV4 DRV44

TXG12
TX_SDI11 SDI DRV1 DRV45
TX_SCK_C SCK DRV2 DRV46
TX_LATCH_C LATCH DRV3 DRV47
TX_EN3 EN DRV4 DRV48

TXG13
TX_SDI12 SDI DRV1 DRV49
TX_SCK_D SCK DRV2 DRV50
TX_LATCH_D LATCH DRV3 DRV51
TX_EN4 EN DRV4 DRV52

TXG14
TX_SDI13 SDI DRV1 DRV53
TX_SCK_D SCK DRV2 DRV54
TX_LATCH_D LATCH DRV3 DRV55
TX_EN4 EN DRV4 DRV56

TXG15
TX_SDI14 SDI DRV1 DRV57
TX_SCK_D SCK DRV2 DRV58
TX_LATCH_D LATCH DRV3 DRV59
TX_EN4 EN DRV4 DRV60

TXG16
TX_SDI15 SDI DRV1 DRV61
TX_SCK_D SCK DRV2 DRV62
TX_LATCH_D LATCH DRV3 DRV63
TX_EN4 EN DRV4 DRV64

TXEN1
TX_EN_FPGA IN OUT TX_EN1

TXEN2
TX_EN_FPGA IN OUT TX_EN2

TXEN3
TX_EN_FPGA IN OUT TX_EN3

TXEN4
TX_EN_FPGA IN OUT TX_EN4

CLKB1
TX_SCK IN OUT1 TX_SCK_A
OUT2 TX_SCK_B

CLKB2
TX_SCK IN OUT1 TX_SCK_C
OUT2 TX_SCK_D

CLKB3
TX_LATCH IN OUT1 TX_LATCH_A
OUT2 TX_LATCH_B

CLKB4
TX_LATCH IN OUT1 TX_LATCH_C
OUT2 TX_LATCH_D

CLKB5
ADC_SCLK IN OUT1 ADC_SCLK_A
OUT2 ADC_SCLK_B

CLKB6
ADC_CONVST IN OUT1 ADC_CONVST_A
OUT2 ADC_CONVST_B

CLKB7
MIC_CLK_IN IN OUT1 MIC_CLK_A
OUT2 MIC_CLK_B

MIC1
MIC_CLK_A CLK DATA MIC_DATA0
3V3_MIC SEL

MIC2
MIC_CLK_A CLK DATA MIC_DATA0
GND SEL

MIC3
MIC_CLK_A CLK DATA MIC_DATA1
3V3_MIC SEL

MIC4
MIC_CLK_A CLK DATA MIC_DATA1
GND SEL

MIC5
MIC_CLK_B CLK DATA MIC_DATA2
3V3_MIC SEL

MIC6
MIC_CLK_B CLK DATA MIC_DATA2
GND SEL

MIC7
MIC_CLK_B CLK DATA MIC_DATA3
3V3_MIC SEL

MIC8
MIC_CLK_B CLK DATA MIC_DATA3
GND SEL

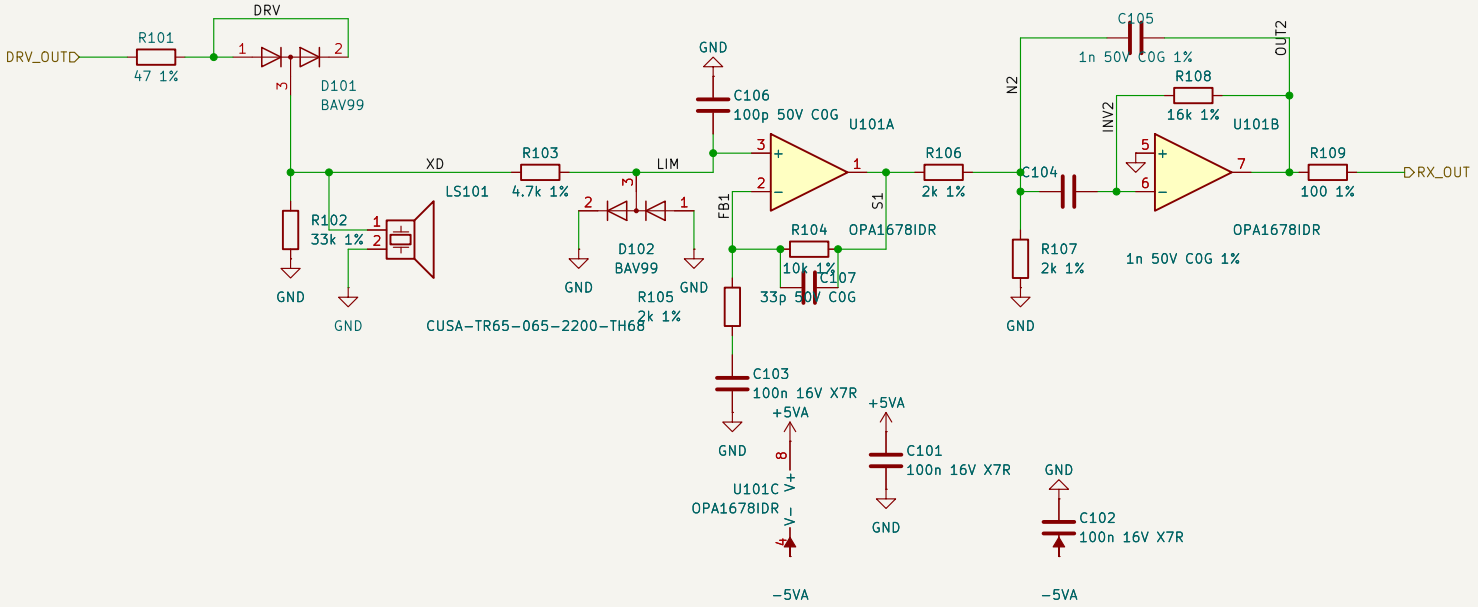
Power

AXU3EGB headers

Sheet: /
File: aesa.kicad_sch

Title: Ultrasonic AESA front end

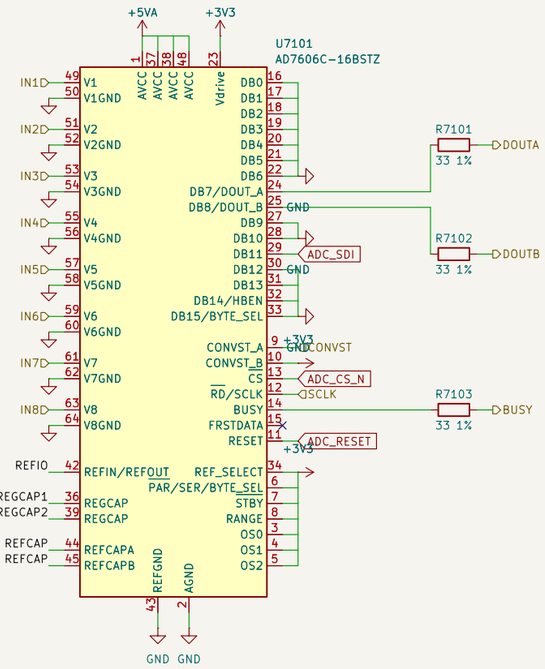
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/110



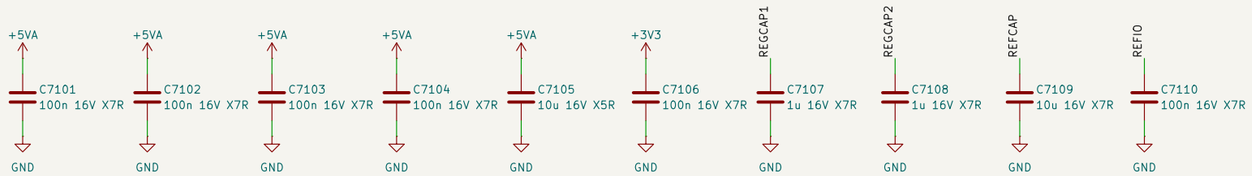
Sheet: /CH01/		
File: channel.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/110

ADC: AD7606C-16 (pin-compatible with AD7606B; symbol keeps AD7606 pin names)

Software mode: OS0-OS2 high. FPGA writes registers over SDI (pin 29, DB11/SDI) at boot:
RANGE_CHx = 0x1 (+-5 V), BANDWIDTH = 0xFF (high bandwidth on all 8 channels), CONFIG DOUT_FORMAT = 2 lines.
Pin 10 is WR on the B/C parts (parallel register writes only): held high. RANGE pin is ignored in software mode.
33 R source resistors on DOUTA/DOUTB/BUSY (ribbon to the FPGA).



Decoupling: C1-C4 at the AVCC pins, C5 bulk, C6 at VDRIVE; REGCAP, REFCAP, REFIN/OUT per datasheet



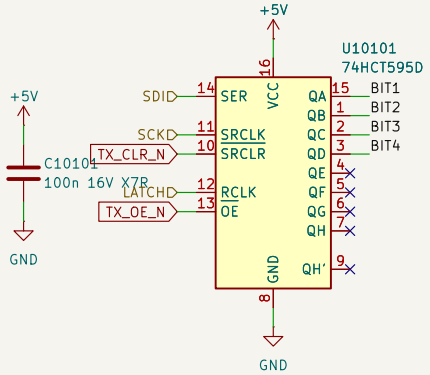
Sheet: /ADC1/
File: adc.kicad_sch

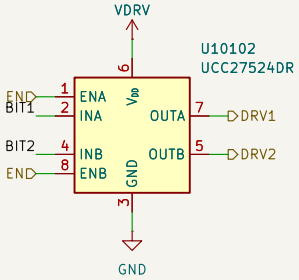
Title:

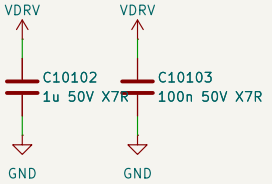
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 66/110

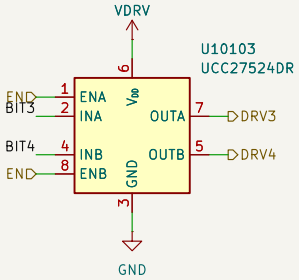
TX group: 4 elements. 74HCT595 QA–QD -> UCC27524 inputs; QE–QH unused (16 data lines, 10 MHz SRCLK)

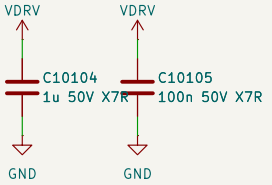
EN from one 74AHCT1G125 per 8 drivers (en_buf). SCK/LATCH from a clk_buf output (<= 4 loads).











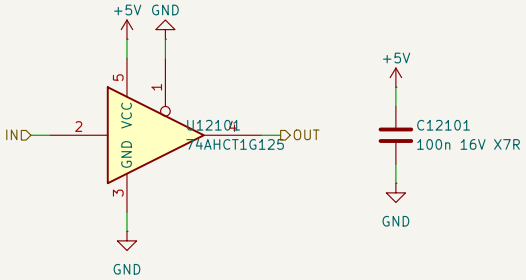
Sheet: /TXG01/
File: tx_group.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 74/110

TX enable buffer: 74AHCT1G125, 3.3 V in (TTL threshold), 5 V out to 8 UCC27524 (16 EN pins)

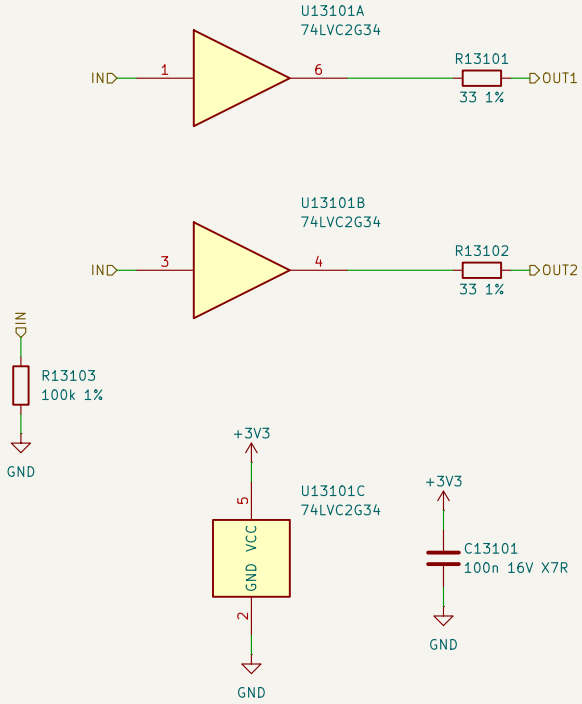


Sheet: /TXEN1/		
File: en_buf.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 90/110

Clock fan-out: 74LVC2G34 at the header, both gates on IN, 33 R series termination per output

Each output drives at most 4 loads (series termination works for a lumped load at the far end).

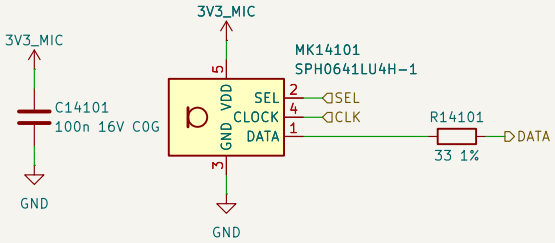
100 k pull-down keeps the clock low while the FPGA is unconfigured.



Sheet: /CLKB1/		
File: clk_buf.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 94/110

MEMS mic SPH0641LU4H-1 (PDM). Two mics share DATA: SEL high on one, low on the other.

Mounted on the back, sound port through a board hole. 3V3_MIC is the ferrite-filtered rail (power sheet).



Sheet: /MIC1/
File: mic.kicad_sch

Title:

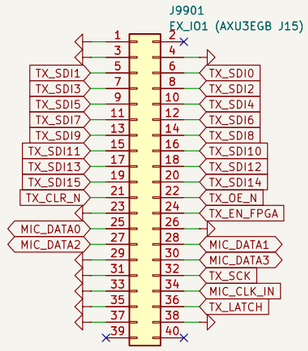
Size: A4 | Date: | **Rev:**

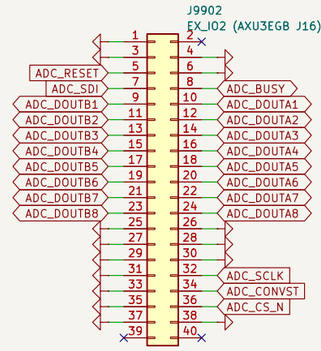
KiCad E.D.A. 10.0.6 | Id: 101/110

AXU3EGB expansion headers (40-pin IDC, 2.54 mm). J1 = board J15 (TX, mics), J2 = board J16 (ADCs).

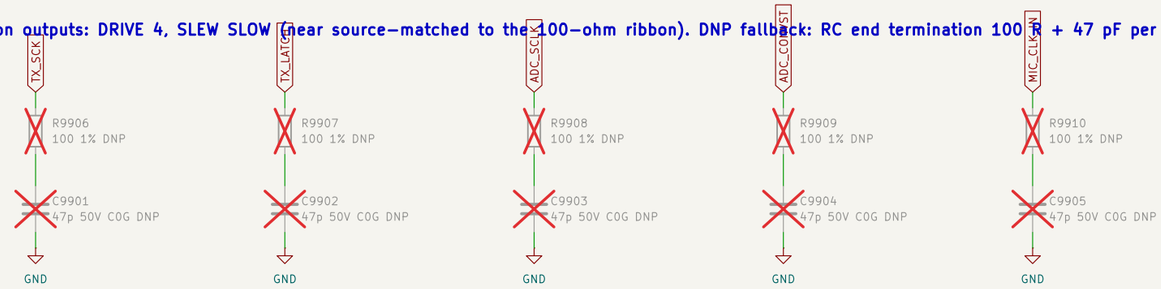
Pin 2 (+5V) and 39/40 (+3.3V) of the FPGA board are left open. Clocks sit between GND conductors (pins 31-37).

Spare FPGA pins tied to GND must stay inputs in the XDC. FPGA outputs: DRIVE 4, SLEW SLOW.

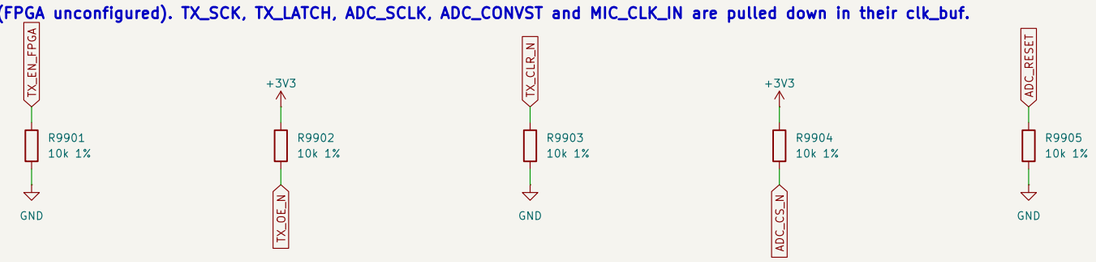




FPGA ribbon outputs: DRIVE 4, SLEW SLOW (near source-matched to the 100-ohm ribbon). DNP fallback: RC end termination 100 R + 47 pF per clock, fit only with DRIVE 8.



Idle pulls (FPGA unconfigured). TX_SCK, TX_LATCH, ADC_SCLK, ADC_CONVST and MIC_CLK_IN are pulled down in their clk_buf.



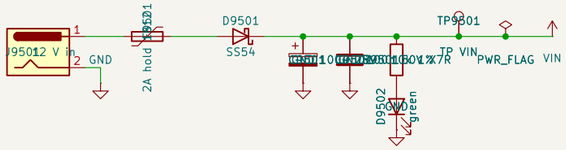
Sheet: /AXU3EGB headers/
File: headers.kicad_sch

Title:

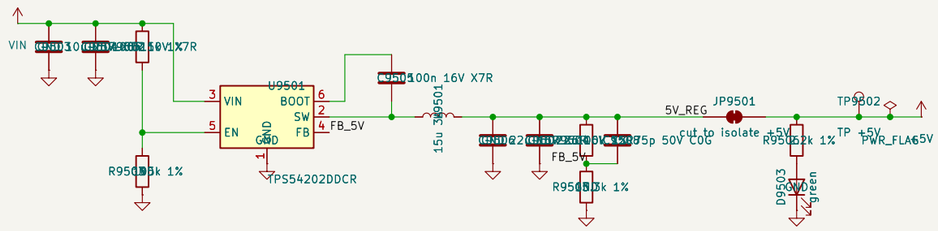
Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 110/110

12 V INPUT (jack, polyfuse, SS54)



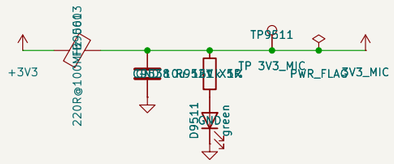
+5V TPS54202 buck (TI ref. design)



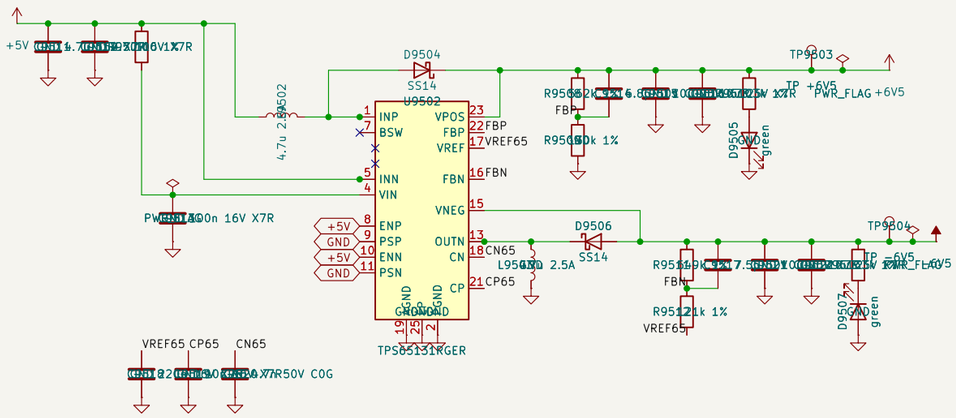
GROUND single plane, test points



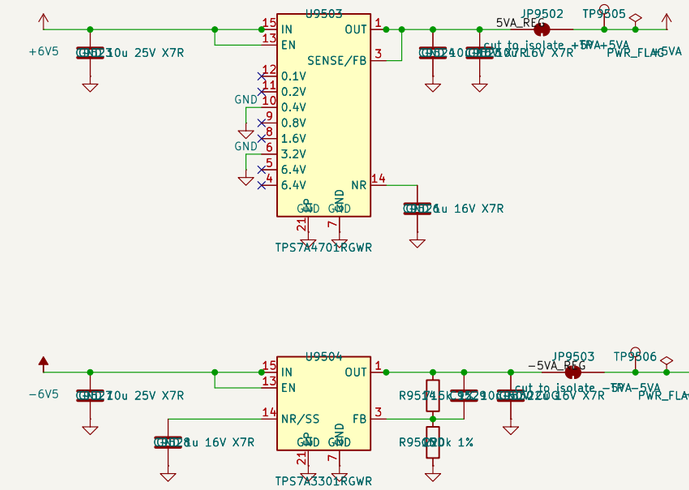
3V3_MIC ferrite filter for the MEMS mics



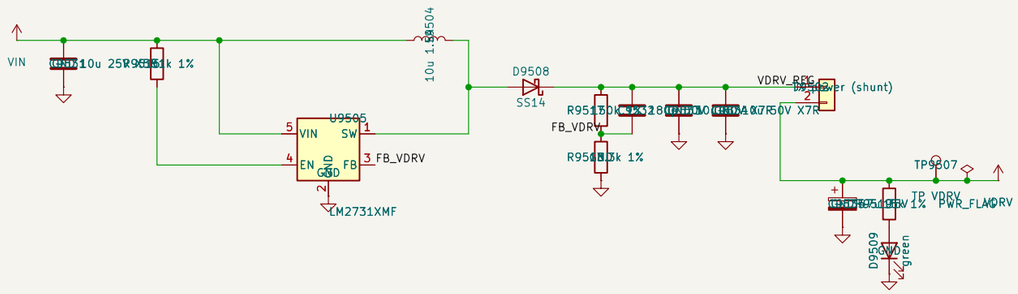
+/−6.5 V PRE-REGULATOR TPS65131 (feeds the LDOs)



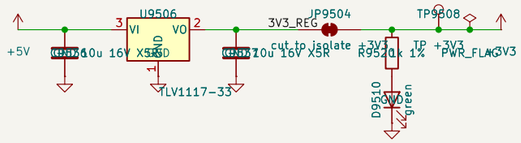
ANALOG RAILS TPS7A4701 +5VA, TPS7A3301 −5VA



VDRV 15 V LM2731X boost; J2 shunt = TX power



+3V3 TLV1117-33



Sheet: /Power/	
File: power.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 109/110